TriMeter

Fair fares for every tricycle ride: a GPS fare meter for Cabanatuan City

BSIT-4C

Mobile Application Technology · Final Case Study

GROUP MEMBERS

John Mathew Ramoso

Neo Arwen De Guzman

Charles Benedict Leodones

Project files

APK, videos and this report

Google Drive folder

Source code

With setup instructions (README)

github.com/mintchoco0624/trimeter-app

01 Project Information

An Android app that meters a tricycle ride with GPS and shows the fair fare from the city ordinance.

TITLE	PLATFORM	BUILT WITH	GROUP
TriMeter	**Android APK v1.5**	**Ionic 9 · Angular 22 · Capacitor 8**	**Ramoso · De Guzman · Leodones**

Install and try it

1 Download

`TriMeter.apk` from the Drive folder (Android 7+).

2 Install

Open the file; allow *Install unknown apps* if asked.

3 Set up

Pick a role and passenger type; allow location *while using the app*.

4 Try

Start Trip, move 100 m+, *End Trip*. Or use *Manual estimate*.

No account, password or API key. Trips stay on the phone. Source: `npm install` → `npx ng serve`; APK: `npx ng build` → `npx cap sync android` → `gradlew assembleDebug`.

02 Problem and Proposed Solution

Cabanatuan is the "Tricycle Capital of the Philippines", yet riders cannot check what a ride should cost.

Problem

300%

Studies report drivers charging up to 300% above the prescribed rate. TODAs apply their own rates and many drivers estimate by feel.

Intended users

Students · Workers · Seniors · PWDs · Tourists

Plus **tricycle drivers**, who get a neutral meter that prevents disputes.

Solution

Measure the real distance with GPS, apply the official fare matrix live, then give a receipt with a breakdown and an overcharge check.

Built now

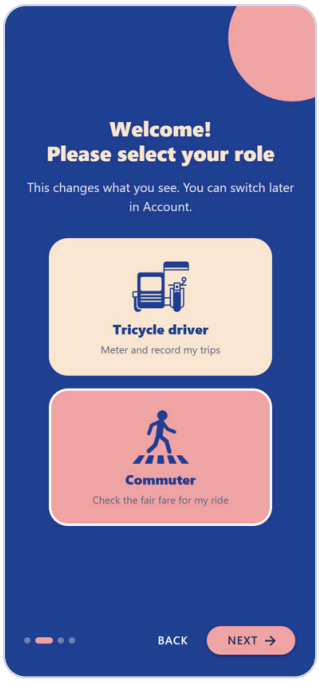
GPS meter · commuter & driver roles · fare preview & calculator · receipts & overcharge check · history · manual estimate · offline map

Future improvements

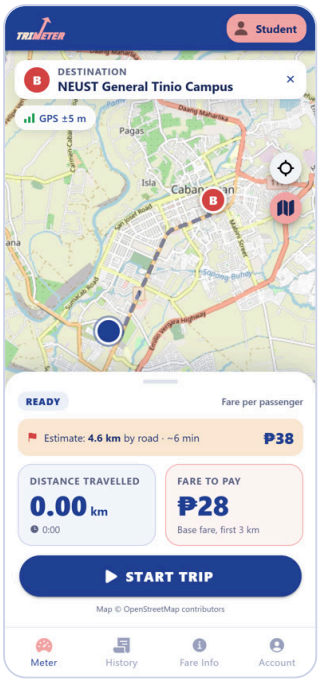
Real accounts · ID-verified discounts · Filipino language · real-time gas prices · driver QR pairing · LGU dashboard

03 Application Features

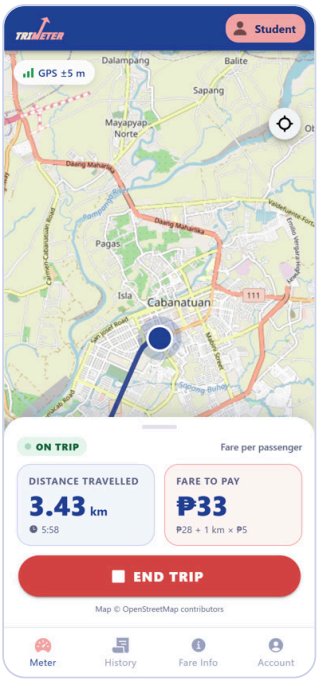
Four tabs: Meter, History, Fare Info and Account. A typical ride takes three taps.



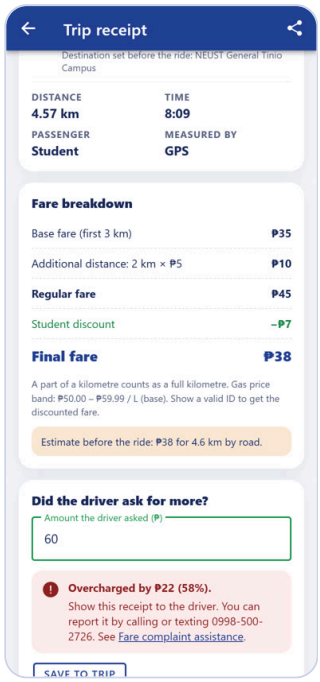
Choose a role
Commuter or driver; passenger type sets the fare.



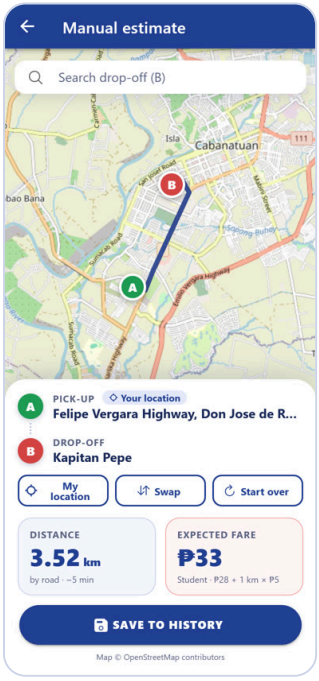
Fare before the ride
Search "Where to?" for road distance and fare.



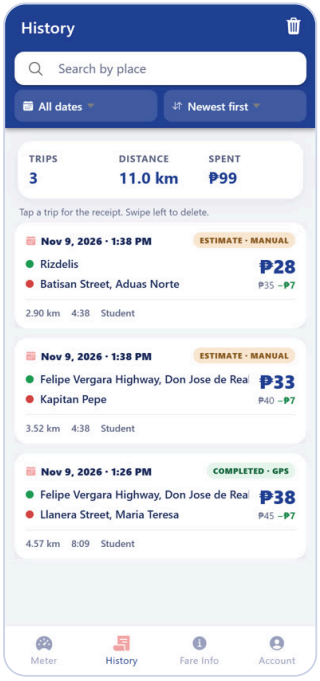
Live meter
Route, distance and fare update as you ride.



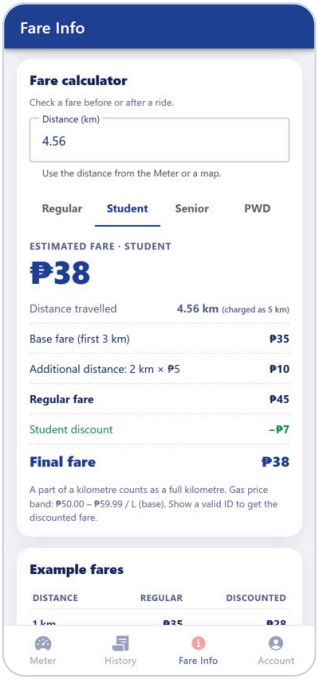
Receipt
Breakdown, overcharge check, share, city hotline.



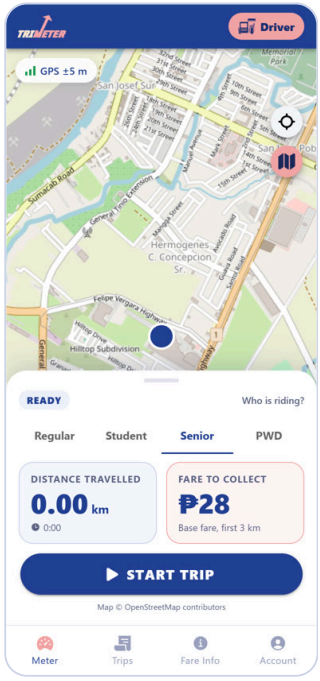
Manual estimate
Starts at your location; pick only the drop-off.



History
Search, filter by date, sort, swipe to delete.



Fare Info
Official guideline, calculator, FAQs, penalties.



Driver mode
Pick the passenger per trip; history = records.

Safety nets: End Trip asks first · trips under 50 m are not charged · a crashed trip can be resumed · the map and meter keep working offline.

04 Technical Implementation

Made with the Ionic CLI: 8 pages, 4 components, 9 services, a route guard and 2 pipes. Pages show data; services do the work.

Pages (UI)
Meter · Estimate · Receipt · History · Fare Info · Account · Onboarding

→

Services (logic)
location · fare · fare-matrix · trip-storage · trip-recovery · routing

→

Device & web
Capacitor Geolocation, Preferences, Haptics, Share · HTTP GET APIs

Capacitor Geolocation: fair GPS

`watchPosition` (high accuracy) sends a fix every 1–2 s. `judgeFix()` rejects:

- accuracy > 35 m
- moves < 5 m (jitter)
- jumps > 30 m/s

Accepted fixes are summed with the haversine formula into Angular signals, so the screen updates by itself.

Fare computation (pure function)

```
fare = base + [km − 3] × rate
```

Distance is rounded to 10 m first, so GPS noise is not billed. Rates follow Ordinance No. 050-2026: ₱35 / ₱28 discounted, +₱5 per km, 11 gas-price bands.

API operations: HttpClient GET

GitHub JSON → saved copy → bundled copy

New fares reach users without a new APK. GETs also call Nominatim (place names) and OSRM (road distance). No API keys.

Data handling: Capacitor Preferences

Trips saved as JSON with full CRUD: add, load, update (amount asked), delete. The trip in progress is saved every 5 s for crash recovery. Map tiles are cached for offline use.

05 Testing and Evaluation

92	11	2	0
unit tests (Vitest)	end-to-end tests (Playwright)	device tests: emulator + real ride	failing tests · lint clean

Feature / task tested	Expected result	Actual result	Status
Role selection (commuter / driver)	Role saved; screens change	As expected	Pass
GPS meter, 4.56 km route (regular)	₱45 (₱35 + 2 km × ₱5)	₱45	Pass
Same route, student passenger	₱38 (₱45 – ₱7 discount)	₱38	Pass
Fare preview before the ride	4.6 km by road, ₱45	As expected	Pass
GPS jitter while standing still	0.00 km added	0.00 km	Pass
Trip with no distance	Not saved as a paid trip	"Discard trip" offered	Pass
Save, search and delete trips	Kept, filtered, deleted after confirm	As expected	Pass
Overcharge check (₱90 asked vs ₱45)	"Overcharged by ₱45 (100%)"	As expected	Pass
Fare data vs. city guideline	All 11 gas bands match	As expected	Pass
App closed mid-trip (emulator)	Trip resumes; student ₱38	4.41 km, ₱38	Pass

Feature / task tested	Expected result	Actual result	Status
Offline map (airplane mode)	Cached map; clear note if missing	As expected	Pass
Real ride on our Android phone (v1.4)	Distance and fare update live	2.77 km, 9:04, GPS ±4–6 m → ₱34	Pass
Tracking with the app in the background	Keeps measuring	Android may pause GPS	Future

Issues and limitations

- GPS drifts near tall buildings and in bad weather.
- The app must stay open; Android can pause GPS in the background.
- The guideline prints ₱4 and ₱5 per km for discounted riders; we confirmed ₱5 is charged.
- No real login: discounts rely on the driver checking an ID.
- Fixed during testing: overlapping map and panels, small A/B pins, a grey map offline.

Recommended improvements

- Foreground service so tracking continues with the screen off.
- Passenger counter for groups of up to 3.
- Filipino language option and a larger-text mode.
- Later: verified ID discounts, driver QR pairing, LGU dashboard.

06 AI Usage Disclosure

Tool used

Claude Code (Anthropic), an AI coding assistant in VS Code.

How it helped

- Scaffolding with the Ionic CLI
- Writing and refactoring the code
- Unit and end-to-end tests
- Our improvement brief; checking fare sources

What we did

- Supplied the Figma design and the city fare guideline
- Reviewed the code and ran all tests
- Tested on our phone; reported issues that were fixed
- Recorded a real ride and the narration in our own voices

We can explain the main code (`location.ts`, `fare.ts`, `fare-matrix.ts`, `trip-storage.ts`). No passwords, private keys or personal data were used.

References

- City Government of Cabanatuan (2026). *Cabanatuan City Tricycle Fare Guidelines*, Ordinansa Blg. 050-2026 (April 14, 2026).
- BetterCabanatuan. *bettercabanatuan-fare*. https://github.com/BetterCabanatuan/bettercabanatuan-fare
- ResearchPublish (2016). *Cabanatuan City: Tricycle Capital of the Philippines*. https://www.researchpublish.com/upload/book/Cabanatuan%20City-3270.pdf
- *Economic Suitability of Electronic Fare Meters for Tricycle Drivers in Cabanatuan City* (2025). ResearchGate.
- Ionic Framework, Capacitor and Angular documentation; OpenStreetMap, Nominatim and OSRM.